

1. A triangle has vertices: A(1,2),B(4,2),C(2,5)

The following transformations are applied sequentially: Scaling by $S_x=2, S_y=3$, Rotation by 90° counter-clockwise about the origin and Translation by $T(3,-2)$.

Write the homogeneous transformation matrix for each transformation. And Determine the transformed coordinates of the triangle.

Marks- 5

2. A cube vertex P(1,2,3) undergoes the following transformations: Translation by (4,-1,2), Scaling by (2,2,3) and Rotation about the x-axis by 180° . Find the composite matrix and Determine the final coordinates of P.

Marks- 5

3. A cubic Bézier curve is defined by control points: $P_0(1,1), P_1(2,5), P_2(6,5), P_3(7,1)$

a) Write the cubic Bézier blending functions.

b) Derive the parametric equation of the curve.

c) Find the coordinates of the curve at $t=0.5$.

d) Show mathematically that the curve passes through P_0 and P_3 .

Marks- 5

4. A Hermit curve is defined using endpoints: $P_0(1,2), P_1(6,4)$ and tangent vectors: $T_0(3,5), T_1(4,-2)$.

a) Write the Hermit basis matrix.

b) Derive the parametric equation of the curve.

c) Find the point on the curve at $t=0.25$.

d) Determine the tangent vector at $t=0.5$.

Marks-5